PHẦN TỬ MITC4

LÝ THUYẾT TẤM REISSNER\MINDLIN

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Nội dung trình bày

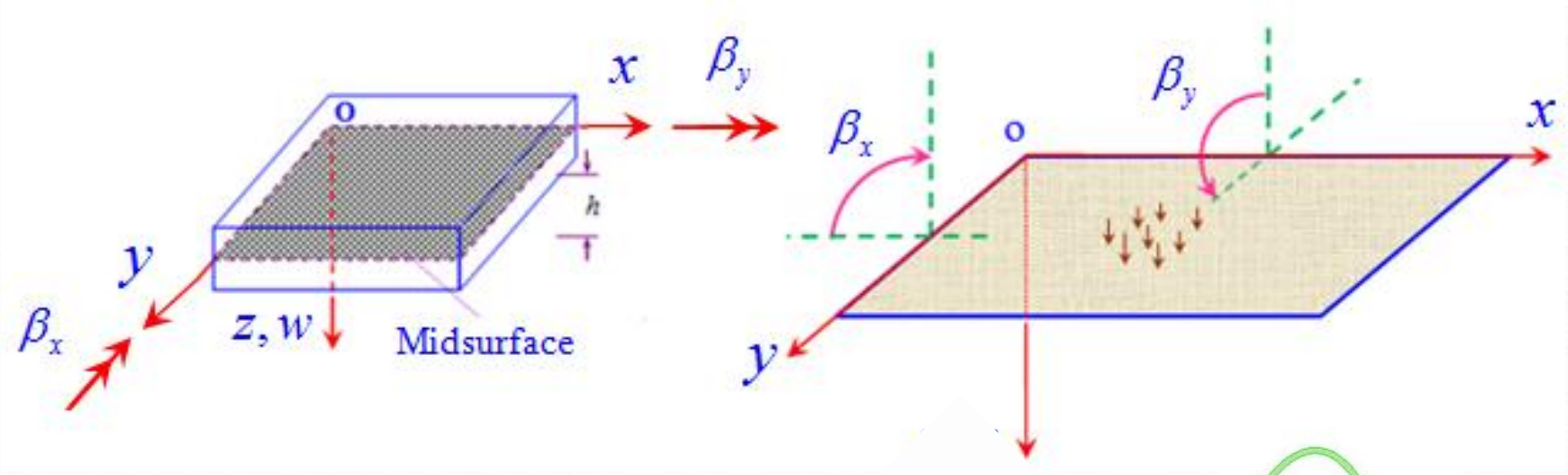


- Quy ước dấu trong tấm Reissner\Mindlin
- Cách thành lập Bb (biến dạng uốn)
- Xử lý ma trận Bs (biến dạng cắt)



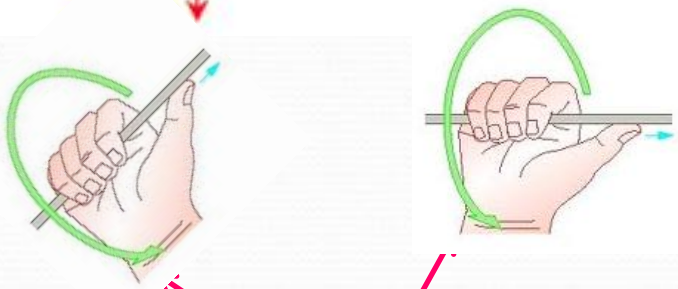
Quy ước dấu

➢ Quy ước dấu trong tấm Reissner\Mindlin



- β_y là góc quay xung quanh trục x
- β_x là góc quay xung quanh trục y

➔ Xác định theo quy tắc bàn tay phải



Chiều quay của β theo độ võng

Thành lập Bb



- Thành lập công thức Bb
  - Biến dạng uốn $u = u_0 + z\beta_x \quad v = v_0 + z\beta_y$

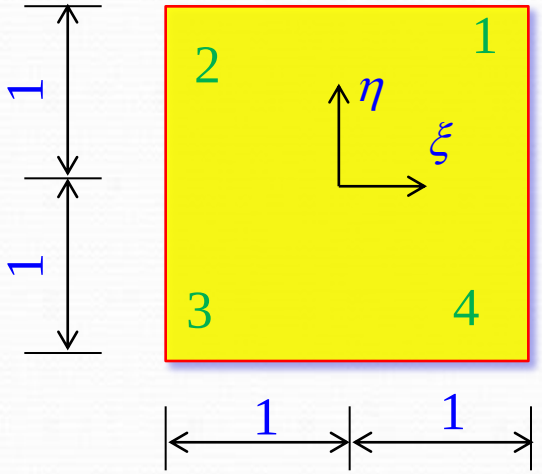
$$\varepsilon_b = \begin{bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial}{\partial x}(u_0 + z\beta_x) \\ \dfrac{\partial v}{\partial y}(v_0 + z\beta_y) \\ \dfrac{\partial}{\partial y}(u_0 + z\beta_x) + \dfrac{\partial}{\partial x}(v_0 + z\beta_y) \end{bmatrix} = z \begin{bmatrix} \dfrac{\partial \beta_x}{\partial x} \\ \dfrac{\partial \beta_y}{\partial y} \\ \dfrac{\partial \beta_x}{\partial y} + \dfrac{\partial \beta_y}{\partial x} \end{bmatrix} = z\kappa$$

$$\kappa = \underbrace{\begin{bmatrix} 0 & \dfrac{\partial}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial}{\partial y} \\ 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix}}_{\mathbf{U}_\beta} = \begin{bmatrix} 0 & \dfrac{\partial}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial}{\partial y} \\ 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \underbrace{\begin{bmatrix} 0 & \dfrac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \dfrac{\partial N_i}{\partial y} \\ 0 & \dfrac{\partial N_i}{\partial y} & \dfrac{\partial N_i}{\partial x} \end{bmatrix}}_{\mathbf{B}^b} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^b \mathbf{d}_\beta$$



Thành lập Bb

➢ Thành lập công thức Bb



$$N_1=\frac{1}{4}(1+\xi)(1+\eta)\quad;\quad N_2=\frac{1}{4}(1-\xi)(1+\eta)$$
$$N_3=\frac{1}{4}(1-\xi)(1-\eta)\quad;\quad N_4=\frac{1}{4}(1+\xi)(1-\eta)$$

$$N_{1,\xi}=0.25(1+\eta)\quad;\quad N_{2,\xi}=-0.25(1+\eta)$$
$$N_{3,\xi}=-0.25(1-\eta)\quad;\quad N_{4,\xi}=0.25(1-\eta)$$
$$N_{1,\eta}=0.25(1+\xi)\quad;\quad N_{2,\eta}=0.25(1-\xi)$$
$$N_{3,\eta}=-0.25(1-\xi)\quad;\quad N_{4,\eta}=-0.25(1+\xi)$$

$$x=\sum_{i=1}^{4}N_i x_i\ \Rightarrow\ x_{,\xi}=\sum_{i=1}^{4}N_{i,\xi}x_i\ ;\ x_{,\eta}=\sum_{i=1}^{4}N_{i,\eta}x_i$$
$$y=\sum_{i=1}^{4}N_i y_i\ \Rightarrow\ y_{,\xi}=\sum_{i=1}^{4}N_{i,\xi}y_i\ ;\ y_{,\eta}=\sum_{i=1}^{4}N_{i,\eta}y_i$$

$$\mathbf{J}=\begin{bmatrix}x_{,\xi} & x_{,\eta}\\ y_{,\xi} & y_{,\eta}\end{bmatrix}=\begin{bmatrix}\sum_{i=1}^{4}N_{i,\xi}x_i & \sum_{i=1}^{4}N_{i,\eta}x_i\\ \sum_{i=1}^{4}N_{i,\xi}y_i & \sum_{i=1}^{4}N_{i,\eta}y_i\end{bmatrix}$$

Thành lập Bb

➢ Thành lập công thức Bb

$$\mathbf{B}^b = \begin{bmatrix} 0 & \frac{\partial N_i}{\partial x} & 0 \\ 0 & 0 & \frac{\partial N_i}{\partial y} \\ 0 & \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{bmatrix}$$

$$\frac{\partial N_i}{\partial x} = \frac{\partial N_i}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial N_i}{\partial \eta}\frac{\partial \eta}{\partial x} = N_{i,\xi}\xi_{,x} + N_{i,\eta}\eta_{,x}$$

$$\frac{\partial N_i}{\partial y} = \frac{\partial N_i}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial N_i}{\partial \eta}\frac{\partial \eta}{\partial y} = N_{i,\xi}\xi_{,y} + N_{i,\eta}\eta_{,y}$$

$$\mathbf{J}^{-1} = \begin{bmatrix} \xi_{,x} & \eta_{,x} \\ \xi_{,y} & \eta_{,y} \end{bmatrix}$$

○ Biến dạng cắt

$$\gamma = \begin{bmatrix} \gamma_{xz} \\ \gamma_{yz} \end{bmatrix} = \begin{bmatrix} w'_x + \beta_x \\ w'_y + \beta_y \end{bmatrix} = \begin{bmatrix} \partial_{,x} & 1 & 0 \\ \partial_{,y} & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} w \\ \beta_x \\ \beta_y \end{bmatrix}}_{\mathbf{U}_\beta} = \begin{bmatrix} \partial_{,x} & 1 & 0 \\ \partial_{,y} & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} N_i & 0 & 0 \\ 0 & N_i & 0 \\ 0 & 0 & N_i \end{bmatrix}}_{\mathbf{N}_\beta} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta}$$

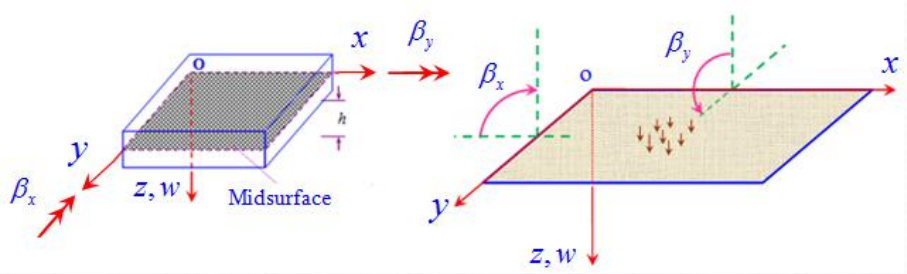
$$= \begin{bmatrix} N_{i,x} & N_i & 0 \\ N_{i,y} & 0 & N_i \end{bmatrix} \underbrace{\begin{bmatrix} w_i \\ \beta_{xi} \\ \beta_{yi} \end{bmatrix}}_{\mathbf{d}_\beta} = \mathbf{B}^s \mathbf{d}_\beta$$

Thành lập Bs

➢ *Khử shear locking*

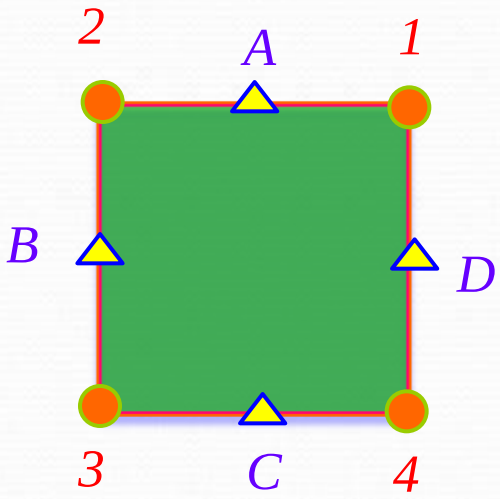
Xấp xỉ lại trường biến dạng

$$w=\sum_{i=1}^{n} N_i w_i \quad ; \quad \beta_x=\sum_{i=1}^{n} N_i \beta_{xi} \quad ; \quad \beta_y=\sum_{i=1}^{n} N_i \beta_{yi}$$



$$\gamma_{xz}=\sum_{i=1}^{n-1} N_i^* \gamma_{xzi} \quad ; \quad \gamma_{yz}=\sum_{i=1}^{n-1} N_i^* \gamma_{yzi}$$

$$\gamma_{yz}=\left(w'_y+\beta_y\right)\Big|_s \; ; \; \gamma_{xz}=\left(w'_x+\beta_x\right)\Big|_s$$



- Phần tử Q4 (MITC4)

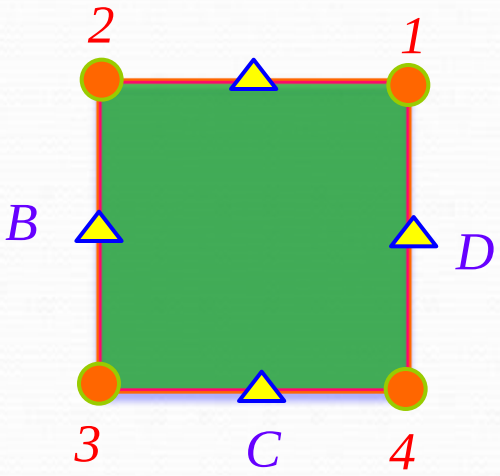
$$\gamma_{xz}=\sum_{i=1}^{n-1} N_i^* \gamma_{xzi}=\underbrace{\frac{1}{2}(1+\eta)}_{N_i^*}\gamma_{xz}^{A}+\frac{1}{2}(1-\eta)\gamma_{xz}^{C}$$

$$\gamma_{yz}=\sum_{i=1}^{n-1} N_i^* \gamma_{yzi}=\frac{1}{2}(1+\xi)\gamma_{yz}^{D}+\frac{1}{2}(1-\xi)\gamma_{yz}^{B}$$

Thành lập Bs

➢ *Khử shear locking*

Xấp xỉ lại trường biến dạng



$$\gamma_{xz}^{A}=\underbrace{\left(\frac{w_1-w_2}{a}\right)}_{w'}+\underbrace{\left(\frac{\beta_{x1}+\beta_{x2}}{2}\right)}_{\beta}\quad;\quad \gamma_{xz}^{C}=\left(\frac{w_4-w_3}{a}\right)+\left(\frac{\beta_{x4}+\beta_{x3}}{2}\right)$$

$$\gamma_{yz}^{D}=\left(\frac{w_1-w_4}{b}\right)-\left(\frac{\beta_{y1}+\beta_{y4}}{2}\right)\quad;\quad \gamma_{yz}^{B}=\left(\frac{w_2-w_3}{b}\right)-\left(\frac{\beta_{y2}+\beta_{y3}}{2}\right)$$

$$\boldsymbol{\gamma}=\mathbf{N}_\gamma\,\boldsymbol{\gamma}\big|_s=\mathbf{N}_\gamma\mathbf{B}_\gamma\,\mathbf{U}=\mathbf{B}_s\mathbf{U}$$

$\mathbf{B}_s$

$$\mathbf{N}_\gamma=\begin{bmatrix}\frac{1}{2}(1+\eta) & \frac{1}{2}(1-\eta) & 0 & 0\\ 0 & 0 & \frac{1}{2}(1+\xi) & \frac{1}{2}(1-\xi)\end{bmatrix}$$

$$\mathbf{B}_\gamma=\begin{bmatrix}\frac{1}{a} & 0 & \frac{1}{2} & -\frac{1}{a} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{a} & 0 & \frac{1}{2} & \frac{1}{a} & 0 & \frac{1}{2}\\ \frac{1}{b} & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{b} & -\frac{1}{2} & 0\\ 0 & 0 & 0 & \frac{1}{b} & -\frac{1}{2} & 0 & -\frac{1}{b} & -\frac{1}{2} & 0 & 0 & 0 & 0\end{bmatrix}$$

THANKS YOUR ATTENTION

